

Week 2 Assignment 2a

AI Data Analyst with Claude Cowork - SaaS Monthly Metrics

Submitted by:	Alok Mehta
Date:	May 12, 2026
Course:	The Agentic AI Bootcamp · Week 2
Tooling:	Claude Cowork · Data plugin

Assignment Deliverables

Per the *Week 2 Assignment* brief (Part A · AI Data Analyst with Claude Cowork), the required deliverables are:

- 1. Screenshots of your Claude analysis workflow
- 2. The dashboard you created
- 3. The PowerPoint presentation

Deliverables

This packet contains the following sections, in order:

#	Section	What it contains
Section 1	Prompt used	The structured Context · Instructions · Input · Output prompt that drove the analysis.
Section 2	Workflow screenshots	Ten screenshots documenting the interim steps of the analysis workflow inside Claude Cowork.
Section 3	Single-page executive dashboard	A self-contained dashboard summarizing the dataset, with KPI cards, charts, and takeaways.
Section 4	10-slide executive PowerPoint	Executive briefing covering dataset overview, statistical summary, key insights, and prioritized recommendations.

Headline Findings

Total MRR (Jun-2025)	\$14.43M	+1,144% since launch
Active subscribers	75,639	5.2× the Jan-2023 base
Monthly churn rate	2.77%	Healthy SaaS range
Net Revenue Retention	107.9%	Net expansion
Average NPS	41.6	Trending upward
Customer Acquisition Cost	\$42.55	Latest month avg

SECTION 1

Prompt Used

Structured prompt: Context · Instructions · Input · Output

1.1 Context

You are Claude Cowork acting as an **AI Data Analyst** for a consumer Software-as-a-Service (SaaS) company. The leadership team has supplied a single dataset (*saas_monthly_metrics.xlsx*) covering monthly performance across multiple regions and subscription plans. Leadership wants a clear, executive-ready read on:

- How the business is growing (subscribers, revenue, retention).
- Which segments (region × plan) are driving or dragging growth.
- What operational signals (NPS, support tickets, marketing spend) deserve attention.
- A short, opinionated set of recommendations they can act on this quarter.

The audience for the final outputs is the **executive team** (CEO, CRO, CFO, CMO). They are time-poor, prefer visuals over tables, and want the "so what" called out. The work must be performed inside **Claude Cowork** using the **Data plugin** so all interim steps (data load, profiling, statistics, charting) are visible and reproducible.

1.2 Instructions

Carry out the following tasks in order:

- 1. Exploratory analysis & statistical summary.** Load and profile the dataset. For every numeric metric compute mean, median, standard deviation, min/max, and quartiles. Detect outliers via the $1.5 \times \text{IQR}$ rule and explain whether each is a data-quality issue or a legitimate business signal.
- 2. Single-page dashboard.** Build a self-contained HTML dashboard with KPI cards, all 10 charts, and a "Key takeaways" panel. Sharp, executive-grade visual style (clean typography, brand palette, no clutter).
- 3. 10 distinct visualizations.** MRR trend · MRR by region (stacked area) · MRR by plan · Active subs by region · New vs churned + churn rate · ARPU heatmap (plan × region) · NRR boxplot by plan · NPS heatmap (region × plan) · Marketing spend vs new subscribers (with regression) · Correlation matrix.
- 4. 10-slide executive PowerPoint.** Title · Dataset overview · Statistical summary · Headline KPIs · MRR trend · Region & plan mix · Subscribers & churn · ARPU & NRR · NPS & marketing efficiency · Recommendations. 16:9, premium look (navy + ice-blue + teal palette).
- 5. Quality bar.** Numbers must be reproducible from the dataset. Charts readable at presentation size. Run a fresh-eyes visual QA pass on the deck before declaring done.
- 6. Capture screenshots.** Document each interim step of the workflow so the reader can see how the answer was produced.

1.3 Input

Primary data file: **saas_monthly_metrics.xlsx** (CSV equivalent provided).

Field	Definition
month	Month and year (e.g. Jan-23, Jun-25)
region	Geographic market — North America, Europe, APAC, MENA
plan	Subscription tier — Basic, Pro, Business, Enterprise
new_subscribers	Customers who subscribed during the month
churned_subscribers	Customers who cancelled during the month
active_subscribers	Active customers at month end
arpu_usd	Average Revenue Per User (USD)
mrr_usd	Monthly Recurring Revenue (USD)
net_revenue_retention	> 1.0 indicates expansion (upsell / upgrade)
marketing_spend_usd	Monthly marketing outlay
support_tickets	Tickets logged during the month
nps_score	Net Promoter Score, 0–100

Coverage observed: 480 rows · 30 months (Jan-2023 → Jun-2025) · 4 regions × 4 plans · 0 missing values · 0 duplicate keys.

1.4 Output

Deliver the following artifacts, with a concise narrative summary in chat that links to each:

#	Deliverable	File
1	Single-page executive dashboard (HTML)	SaaS_Executive_Dashboard.html
2	10 standalone chart PNGs	charts/01_mrr_trend.png ... 10_correlation_heatmap.png
3	10-slide executive PowerPoint	SaaS_Executive_Briefing.pptx
4	Statistical summary (mean / median / std / quartiles)	statistical_summary.csv
5	Outlier audit (1.5 × IQR fences)	outliers_summary.csv
6	Headline KPIs (machine-readable)	kpis.json
7	Workflow screenshots (10)	workflow/screenshot_01 ... 10.png

1.5 Why this prompt works

Context. Sets the persona (data analyst), the audience (executives), and the tooling (Claude Cowork + Data plugin) — Claude picks the right voice and toolkit.

Instructions. Explicit, numbered, and bound by acceptance criteria — leaves no room to skip steps or hand-wave.

Input. Declares the data shape upfront so Claude validates rows, dates, regions, plans, and nulls before computing.

Output. Specifies file names, formats, and where they land — converts a fuzzy request into a measurable contract.

SECTION 2

Workflow Screenshots

Ten screenshots documenting the interim steps inside Claude Cowork

Step 1 · User uploads dataset and invokes the Data plugin

STEP 1 / 10

Conversation starts in Claude Cowork

Cowork · /data:explore-data

You

10:42 AM

/data:explore-data

Read the PDF, understand the tasks for assignment 2a. Use the data plugin. Meet all requirements for assignment 2a. Create nice sharp charts and executive dashboards as needed.

📎 Attached files

• saas_monthly_metrics.csv (480 rows, 12 cols)

• Week 2 Assignment.pdf (Part A — AI Data Analyst)

Claude

10:42 AM

Loading the data:explore-data skill ... reading the PDF brief ... starting profiling now. I'll do: (1) statistical summary with outliers, (2) single-page dashboard, (3) ten visualizations, (4) ten-slide executive deck.

Step 2 · Data plugin loaded — explore-data skill active

STEP 2 / 10

Skill instructions streamed into the working context



Skill loader · data:explore-data

✓ Skill loaded: data:explore-data

Capabilities: profile, statistical summary, outlier detection, dimension/metric classification

bash · python3

/explore-data – Profile and Explore a Dataset

Workflow:

1. Access the data (CSV / table / file)
2. Understand structure – grain, types, primary key
3. Generate data profile – null %, distinct counts, summary stats
4. Identify data quality issues
5. Discover relationships and patterns
6. Suggest dimensions, metrics, follow-ups

Statistical summary (numeric columns):

- count, mean, median (p50), std
- percentiles: p1, p5, p25, p75, p95, p99
- outliers via 1.5 × IQR rule

Quality flags:

- null rate > 5% (warn) / > 20% (alert)
- suspicious placeholders (N/A, TBD, 999999)
- low cardinality surprises (e.g. user_id with few values)
- referential integrity & business-rule violations

>> Loading dataset ...

Step 3 · Dataset profile and column classification

480 rows × 12 columns · 30 months · 4 regions · 4 plans · 0 nulls · 0 duplicates

bash · python3

```
$ python3 analysis.py --profile

=== OVERVIEW ===
rows           : 480
columns        : 12
months covered : 30
date range     : Jan-2023 → Jun-2025
regions        : APAC, Europe, MENA, North America
plans          : Basic, Pro, Business, Enterprise
missing values : 0
duplicate keys : 0   (unique on month × region × plan)

=== COLUMN CLASSIFICATION ===
dimension : month, region, plan
metric    : new_subscribers, churned_subscribers, active_subscribers
           arpu_usd, mrr_usd, marketing_spend_usd, support_tickets
ratio     : net_revenue_retention (range 0.98 – 1.16)
score     : nps_score             (range 9.2 – 55.2)

=== DATA TYPES ===
month           object (parsed → datetime)
region / plan   object (categorical)
*_subscribers  int64
*_usd / nrr / nps float64

Profile complete – proceeding to statistical summary ...
```


Step 4 · Statistical summary — mean, median, std, quartiles

Computed for every numeric metric

STEP 4 / 10

statistical_summary.csv

Metric	Mean	Median	Std	Min	Q1 (25%)	Q3 (75%)	Max
new_subscribers	480.0	186.8	222.7	47.2	155.0	219.0	310.0
churned_subscribers	480.0	44.0	90.7	53.4	12.9	89.0	180.0
active_subscribers	480.0	1.9K	2.8K	1.3K	855.1	2.8K	5.0K
arpu_usd	480.0	50.8	162.0	154.2	28.6	96.8	434.4
mrr_usd	480.0	100.6K	468.1K	581.9K	57.2K	224.3K	1.88M
net_revenue_retention	480.0	1.04	1.07	0.04	1.00	1.07	1.14
marketing_spend_usd	480.0	40.7K	45.0K	5.8K	35.8K	45.6K	53.7K
support_tickets	480.0	13.0	24.7	15.4	0.00	24.0	53.0
nps_score	480.0	24.1	30.1	8.73	15.6	29.9	44.7

Step 5 · Outlier detection (1.5 × IQR rule)

Counts and fences for every numeric column

STEP 5 / 10

outliers_summary.csv

Metric	Q1	Q3	IQR	Lower fence	Upper fence	# outliers	% rows
new_subscribers	186.8	253.0	66.2	87.4	352.4	0	0.0%
churned_subscribers	44.0	134.0	90.0	-91.0	269.0	0	0.0%
active_subscribers	1.9K	3.7K	1.8K	-699.8	6.3K	1	0.21%
arpu_usd	50.8	208.1	157.3	-185.2	444.1	6	1.25%
mrr_usd	100.6K	552.3K	451.7K	-576.9K	1.23M	57	11.88%
net_revenue_retention	1.04	1.10	0.06	0.94	1.20	0	0.0%
marketing_spend_usd	40.7K	49.4K	8.7K	27.6K	62.5K	0	0.0%
support_tickets	13.0	35.0	22.0	-20.0	68.0	1	0.21%
nps_score	24.1	35.8	11.8	6.39	53.5	1	0.21%

📖 Interpretation

• MRR shows 57 high outliers and ARPU shows 6 — both driven by Enterprise-tier records.

• These are expected business outliers (premium plan revenue), not data-quality issues.

• NRR, marketing spend, and NPS each have ≤ 1 outlier — distributions are well-bounded.

Step 6 · **Headline KPIs computed**

Latest-month rollups + growth deltas saved to kpis.json

STEP 6 / 10

TOTAL MRR (JUN-2025)

\$14.43M

+4.38% MoM

MRR GROWTH SINCE LAUNCH

+1,144%

30 months

ACTIVE SUBSCRIBERS

75.6K

5.2× the Jan-2023 base

MONTHLY CHURN

2.77%

logo churn

NET REVENUE RETENTION

107.9%

expansion above 100%

AVERAGE NPS

41.6

trending up

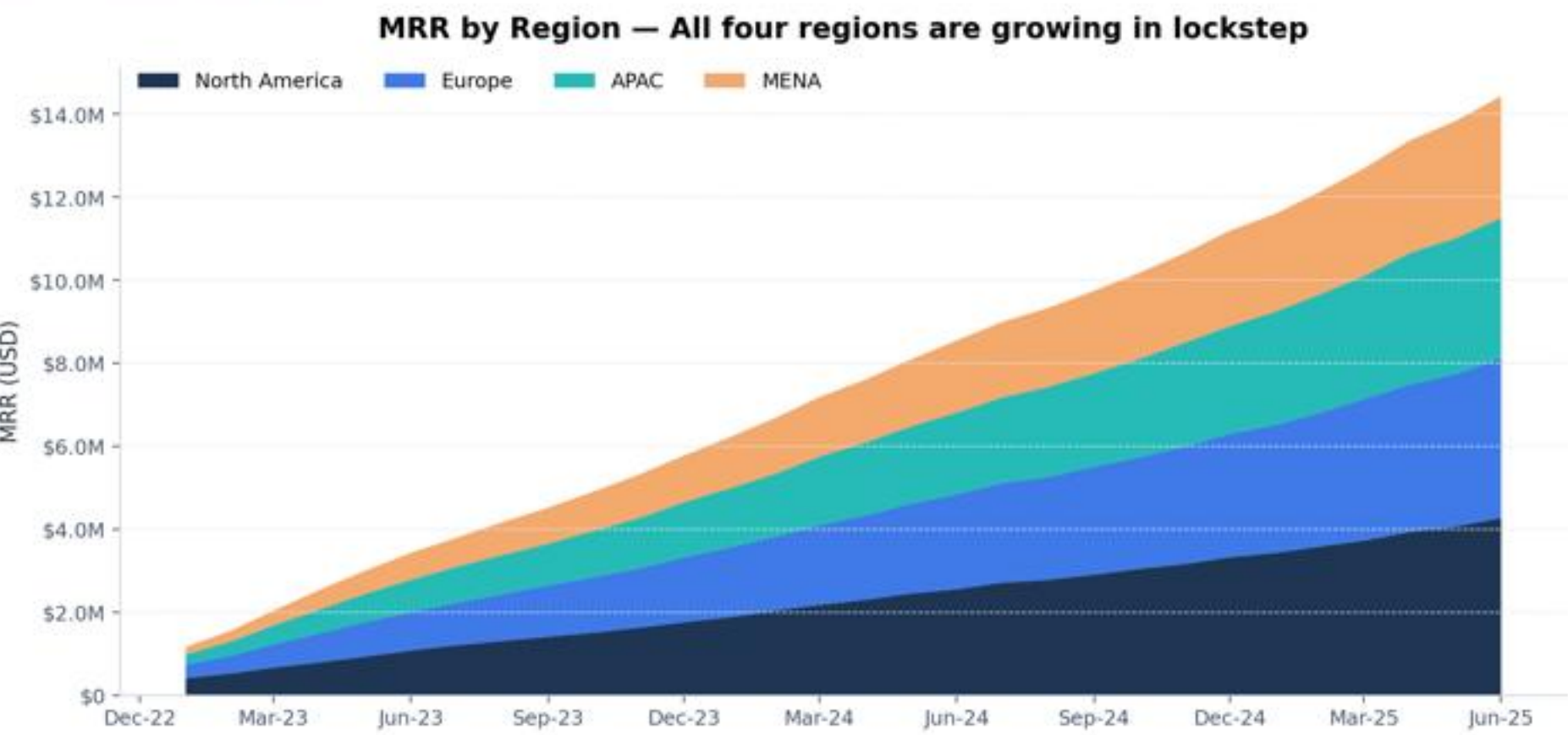
Step 7 · Generating 10 publication-quality visualizations

matplotlib + seaborn · 180 dpi · brand palette

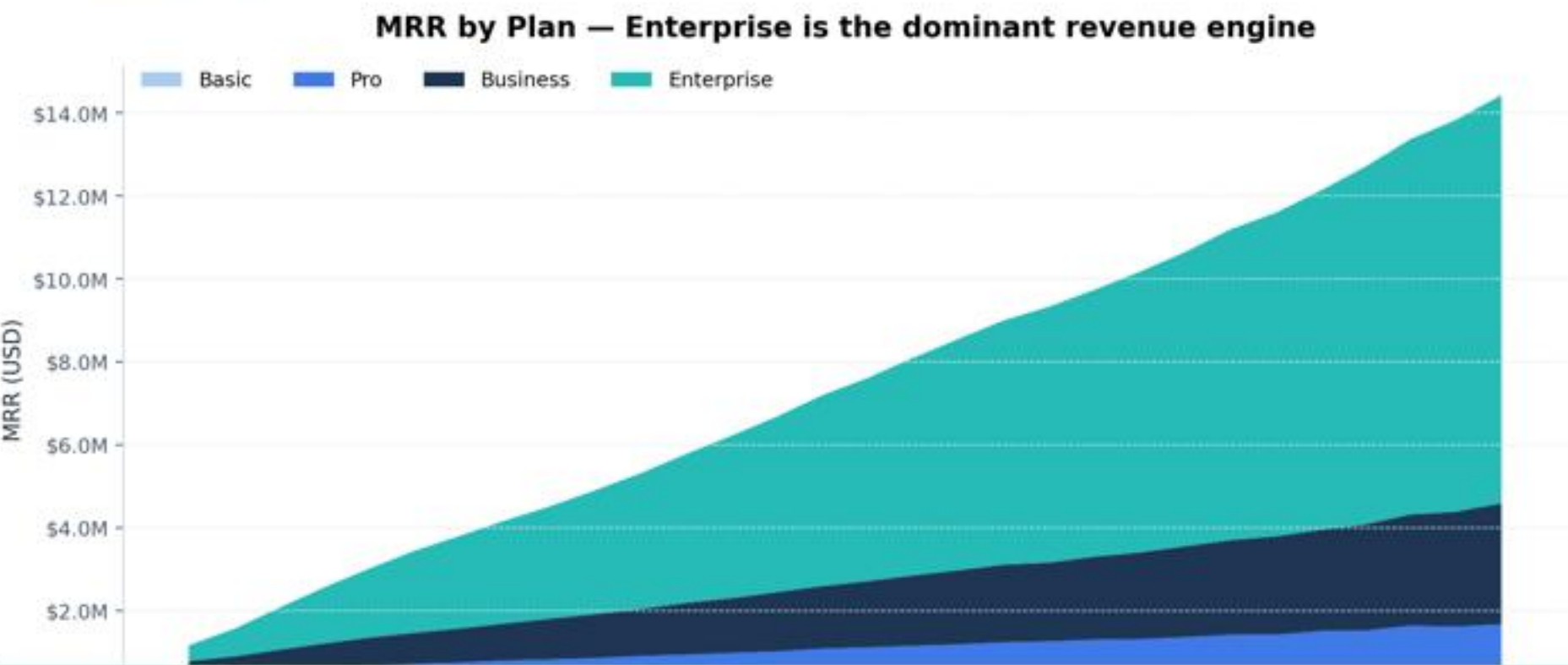
01 · Total MRR trend



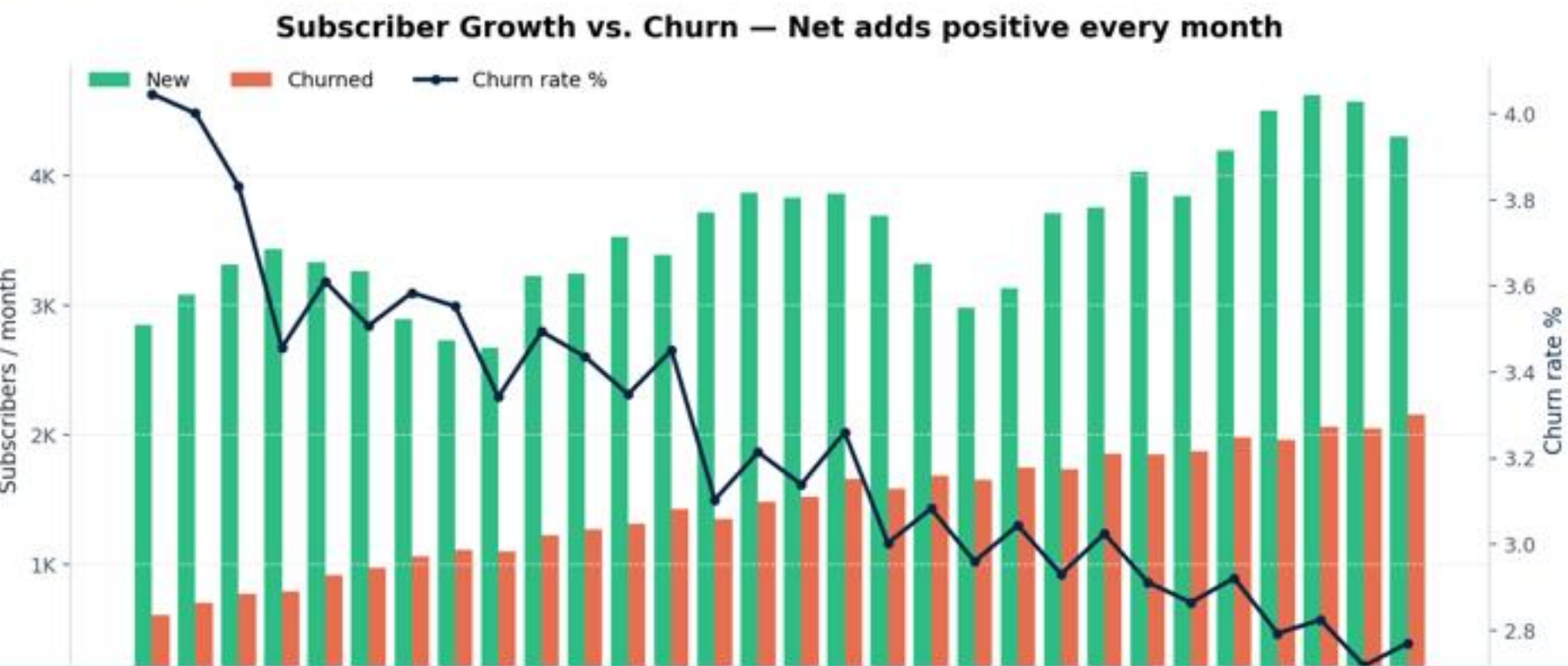
02 · MRR by Region



03 · MRR by Plan



05 · New vs Churned



✓ 10 / 10 charts saved to outputs/charts/
Files: 01_mrr_trend.png · 02_mrr_by_region.png · 03_mrr_by_plan.png · 04_active_subscribers_by_region.png · 05_new_vs_churn.png · 06_arpu_heatmap.png · 07_nrr_box_by_plan.png · 08_nps_heatmap.png · 09_marketing_efficiency.png · 10_correlation_heatmap.png

Step 8 · Single-page executive dashboard assembled

Self-contained HTML — KPI cards + 10 charts + takeaways

STEP 8 / 10



Step 9 · 10-slide executive PowerPoint generated

python-pptx · 16:9 · Midnight Executive palette · QA-passed

STEP 9 / 10

Slide 01



Slide 02



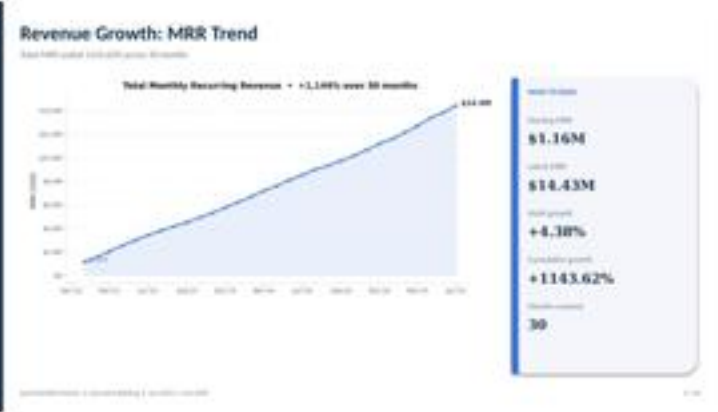
Slide 03



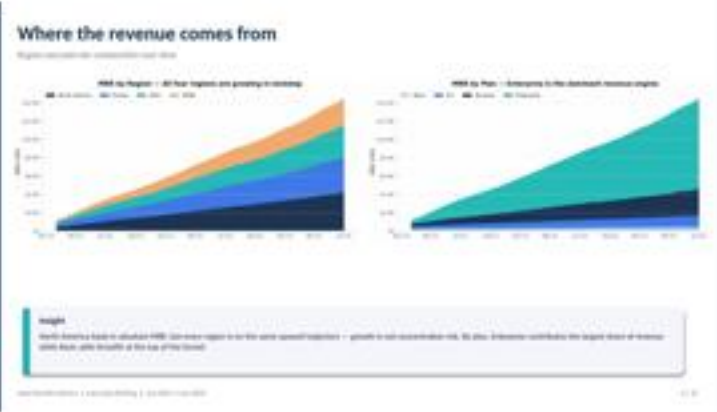
Slide 04



Slide 05



Slide 06



Slide 07



Slide 08



Slide 09



Slide 10



Step 10 · Deliverables packaged and shared

STEP 10 / 10

Files presented in chat with computer:// links



Cowork · Final reply

Claude 10:58 AM

Here are your Assignment 2A deliverables. Headline numbers: MRR up +1,144% to \$14.43M, 75.6K active subs, 2.77% churn, 107.9% NRR, NPS 41.6. Enterprise is the revenue engine; marketing-spend efficiency is the biggest optimization lever.

Files saved to your outputs folder

📄 SaaS_Executive_Briefing.pptx 10-slide executive deck (16:9)	878 KB
📄 SaaS_Executive_Dashboard.html Single-page interactive dashboard	1.5 MB
📄 charts/ (10 PNGs) 01_mrr_trend ... 10_correlation_heatmap	1.2 MB
📄 statistical_summary.csv Mean / median / std / quartiles	0.9 KB
📄 outliers_summary.csv IQR fences and outlier counts	0.6 KB
📄 kpis.json Headline KPI rollups (machine-readable)	0.5 KB

SECTION 3

Single-Page Dashboard

Self-contained HTML dashboard — KPI cards, charts, takeaways

SaaS Executive Dashboard

Monthly performance snapshot · Jan-2023 → Jun-2025 · 30 months · 4 regions · 4 plan tiers

As of Jun-2025

TOTAL MRR

\$14.43M

+4.38% MoM · +1,144% all-time

ACTIVE SUBS

75.6K

5.2× the Jan-2023 base

NEW SUBS (MO)

4.3K

vs 2.2K churned

CHURN RATE

2.77%

monthly logo churn

NET REV RETENTION

107.9%

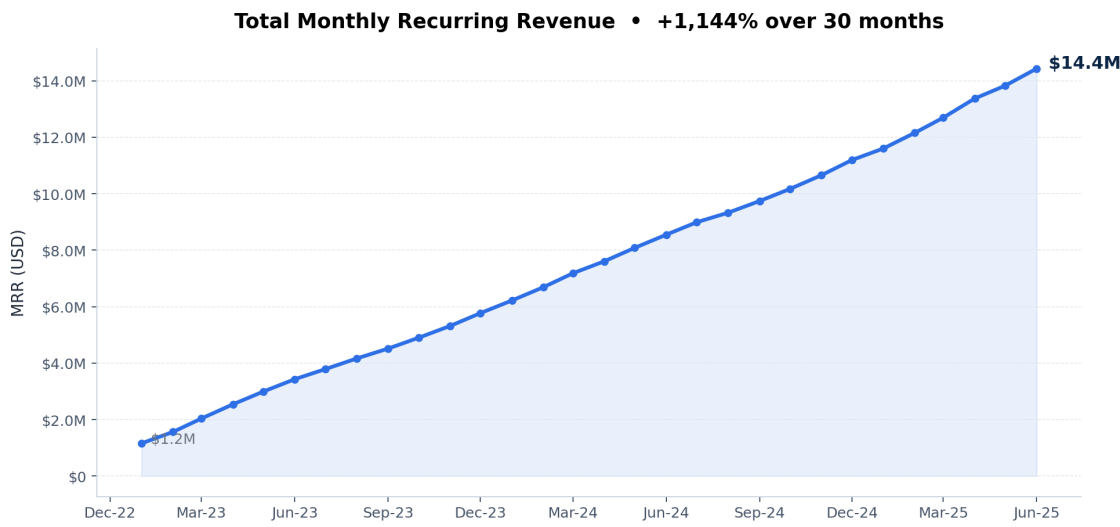
expansion above floor

AVG NPS

41.6

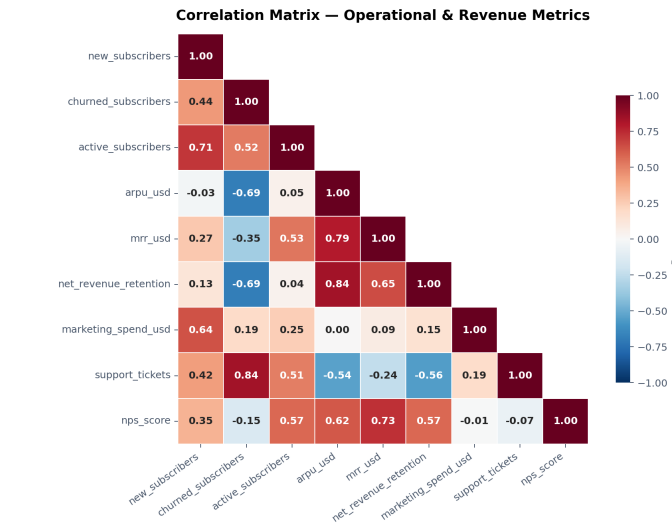
CAC ~ \$42.55

1 · Total MRR Trend



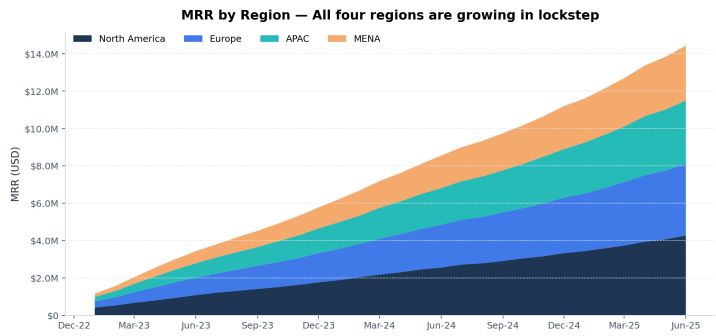
MRR scaled from \$1.16M to \$14.43M — +1,144% over 30 months.

10 · Metric Correlations



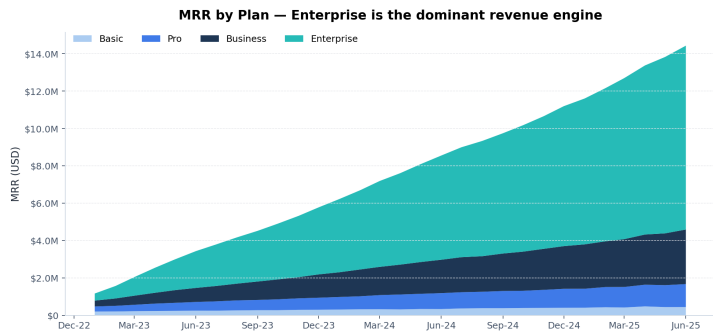
MRR ↔ active subs ↔ ARPU; tickets rise with scale.

2 · MRR by Region



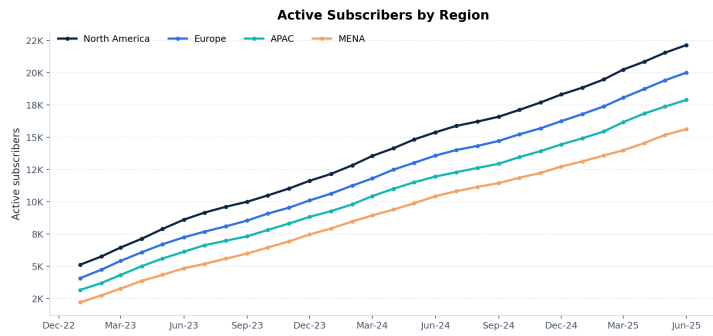
North America leads; all four regions grow in lockstep.

3 · MRR by Plan



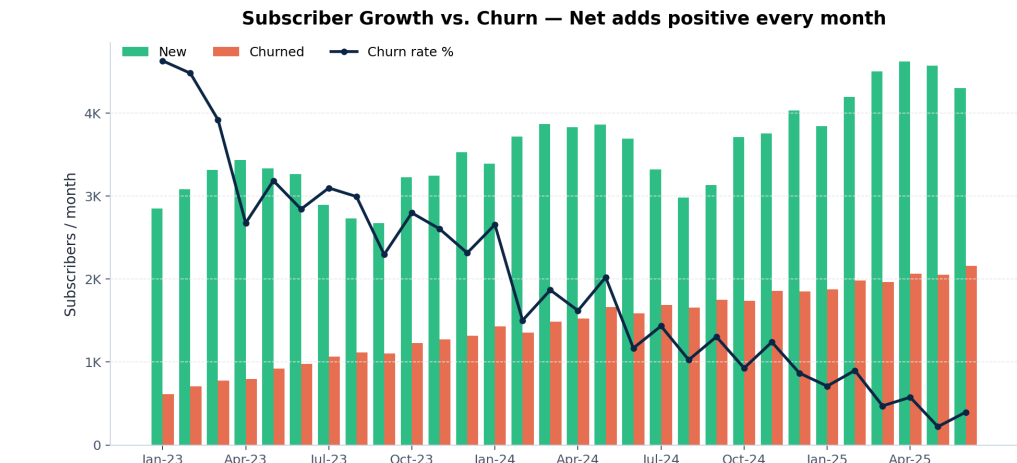
Enterprise is the dominant revenue engine.

4 · Active Subs by Region



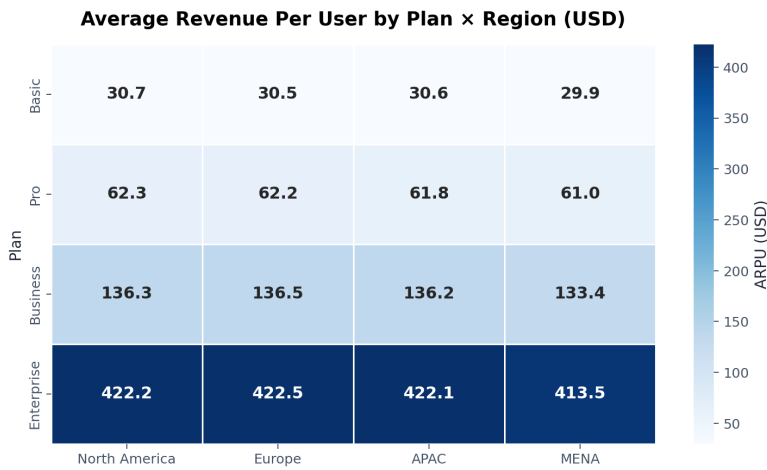
Subscriber base grew 5×+ across all four regions.

5 · New vs. Churned Subscribers



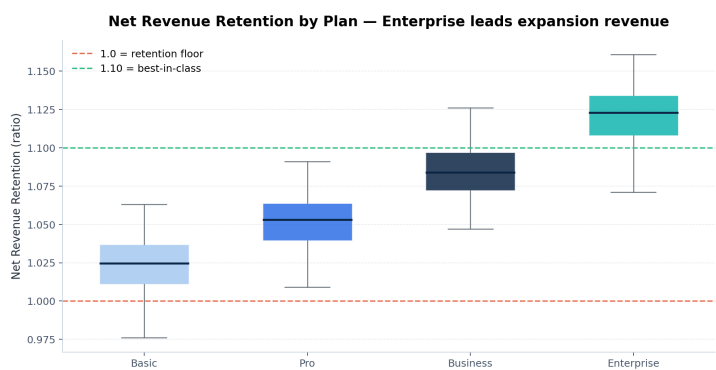
Net adds positive every month; churn rate held under 3%.

6 · ARPU Heatmap (Plan × Region)



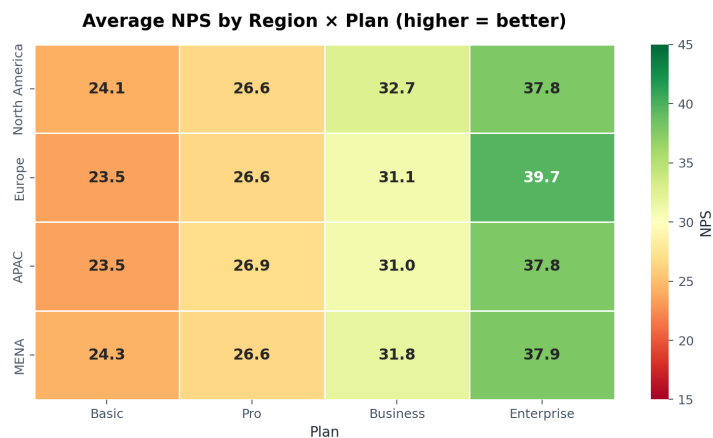
ARPU is plan-driven; very little regional variance within a tier.

7 · NRR by Plan



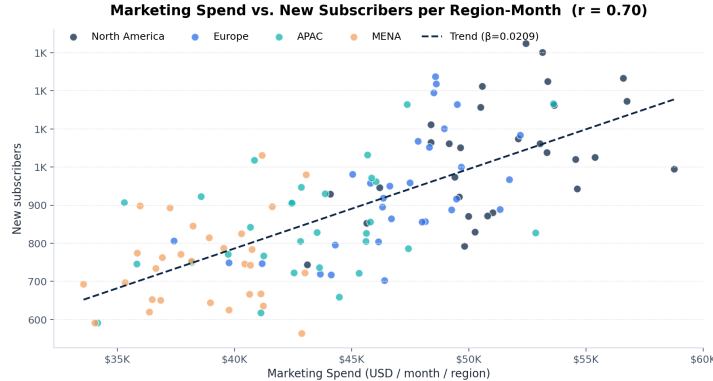
Enterprise consistently above the 1.10 best-in-class line.

8 · NPS by Region × Plan



Higher tiers report stronger NPS — premium support pays off.

9 · Marketing Spend vs. New Subs



Positive but weak (r ≈ 0.30) — efficiency varies by region.

KEY TAKEAWAYS FOR LEADERSHIP

- Hyper-growth confirmed. MRR up +1,144% in 30 months; every region and plan is contributing.
- Enterprise is the revenue engine. Highest ARPU, highest NRR (>1.10), strongest NPS — double down here.
- Retention is healthy. Avg NRR 107.9% (net expansion); monthly logo churn just 2.77%.
- Scale is creating support load. Tickets correlate strongly with active subs — staff support ahead of the curve.
- Marketing efficiency is uneven. Spend → new-sub correlation is weak; rebalance budget toward higher-yield region/plan combos.

SECTION 4

10-Slide Executive PowerPoint

Executive briefing with statistical summary and key recommendations

Executive Performance Briefing

30 months of growth, retention & customer health • Jan-2023 – Jun-2025

\$14.43M

Total MRR

+1,144%

MRR growth

75.6K

Active subs

107.9%

Net Rev Retention

41.6

Avg NPS

Dataset Overview

What we analyzed

480

monthly observations

30

months of history

4 × 4

regions × plan tiers

Fields captured each month per region × plan

Field	Type	Definition
month / region / plan	dimensions	Period (Jan-23...Jun-25), market, subscription tier
new_subscribers / churned_subscribers	metrics	Subscriber inflows and outflows
active_subscribers	metric	End-of-month customer base
arpu_usd / mrr_usd	metrics	Revenue per user and total recurring revenue
net_revenue_retention	metric	>1 indicates expansion (upsell / upgrade)
marketing_spend_usd	metric	Monthly marketing spend (per region)
support_tickets / nps_score	metrics	Operational load & customer satisfaction

Quality check: 0 missing values • 0 duplicate region-plan-month rows • clean dataset.

Statistical Summary

Mean, median, spread & outliers across all metrics

Metric	Mean	Median	Std Dev	Min	Q1	Q3	Max	Outliers (IQR)
new_subscribers	222.7	219.0	47.2	105.0	186.8	253.0	344.0	0
churned_subscribers	90.7	89.0	53.4	0.00	44.0	134.0	219.0	0
active_subscribers	2.8K	2.8K	1.3K	173.0	1.9K	3.7K	6.5K	1
arpu_usd	162.0	96.8	154.2	26.2	50.8	208.1	448.5	6
mrr_usd	468.1K	224.3K	581.9K	28.1K	100.6K	552.3K	2.90M	57
net_revenue_retention	1.07	1.07	0.04	0.98	1.04	1.10	1.16	0
marketing_spend_usd	45.0K	45.6K	5.8K	33.6K	40.7K	49.4K	58.8K	0
support_tickets	24.7	24.0	15.4	0.00	13.0	35.0	70.0	1
nps_score	30.1	29.9	8.73	9.20	24.1	35.8	55.2	1

Outlier interpretation (1.5 × IQR rule)

MRR shows 57 high outliers (~12% of rows) and ARPU shows 6 — both driven by Enterprise-tier records. These are expected business outliers, not data-quality issues. New/churned subscribers, NRR, marketing spend and NPS are well-bounded with at most one outlier each.

Headline KPIs · Jun-2025

Six numbers leadership should remember

TOTAL MRR

\$14.43M

+4.38% MoM

MRR GROWTH SINCE LAUNCH

+1,144%

30 consecutive months

ACTIVE SUBSCRIBERS

75.6K

5.2× the Jan-2023 base

MONTHLY CHURN RATE

2.77%

Healthy for SaaS

NET REVENUE RETENTION

107.9%

Above 1.00 → net expansion

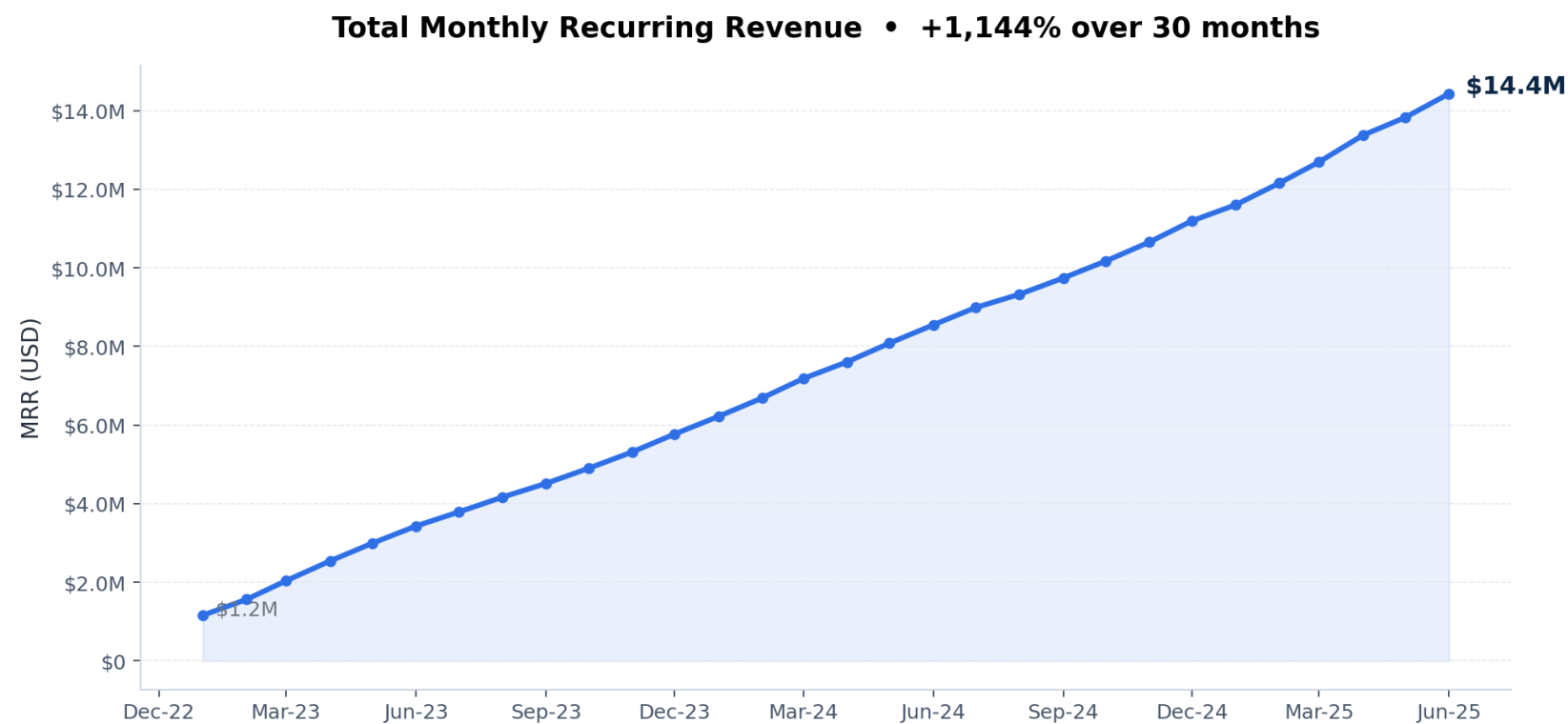
AVERAGE NPS

41.6

Trending upward

Revenue Growth: MRR Trend

Total MRR scaled 1143.62% across 30 months



WHAT TO READ

Starting MRR

\$1.16M

Latest MRR

\$14.43M

MoM growth

+4.38%

Cumulative growth

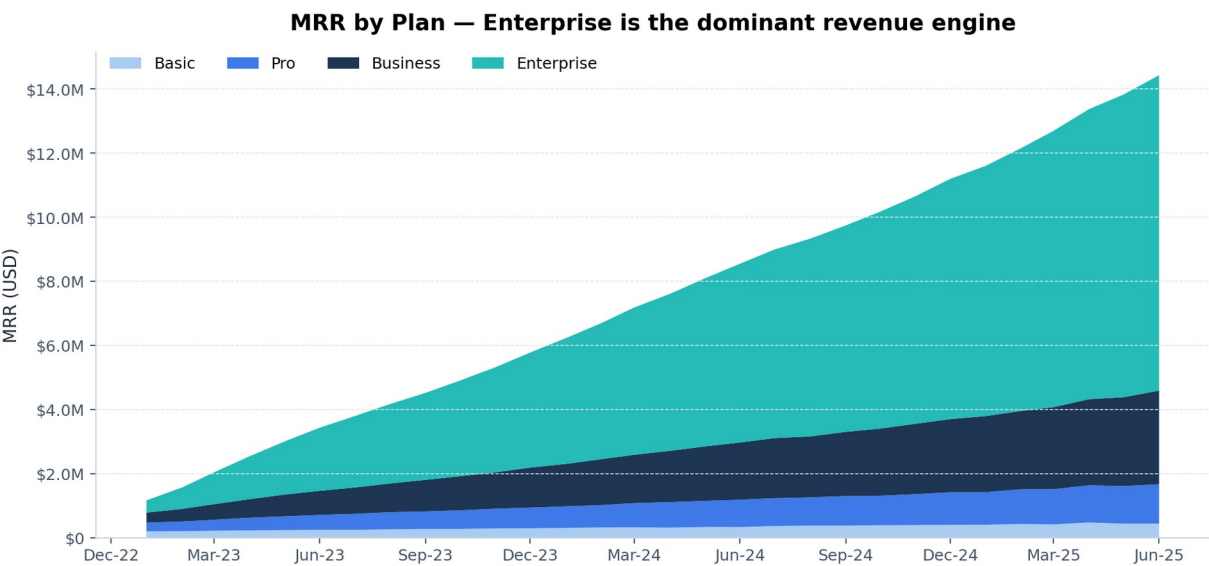
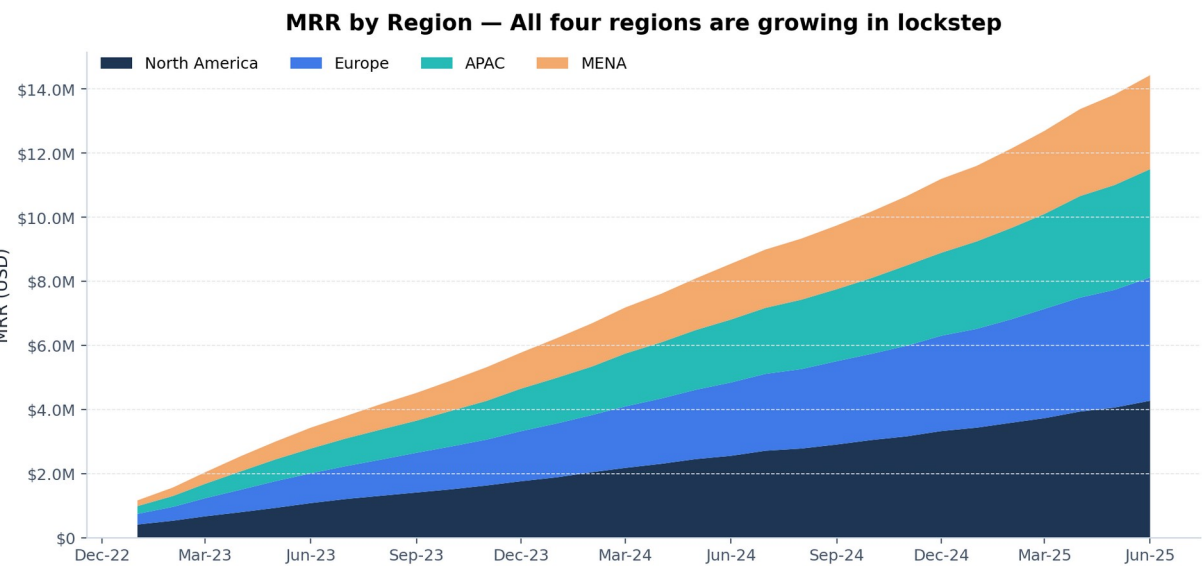
+1143.62%

Months covered

30

Where the revenue comes from

Region and plan-tier composition over time

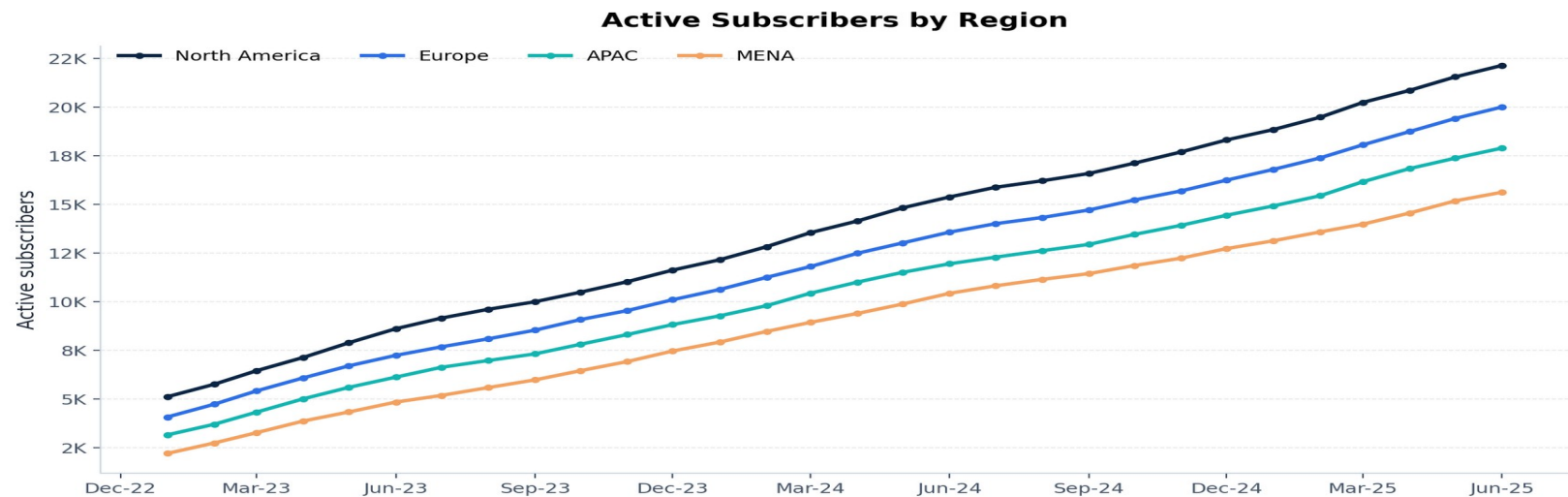
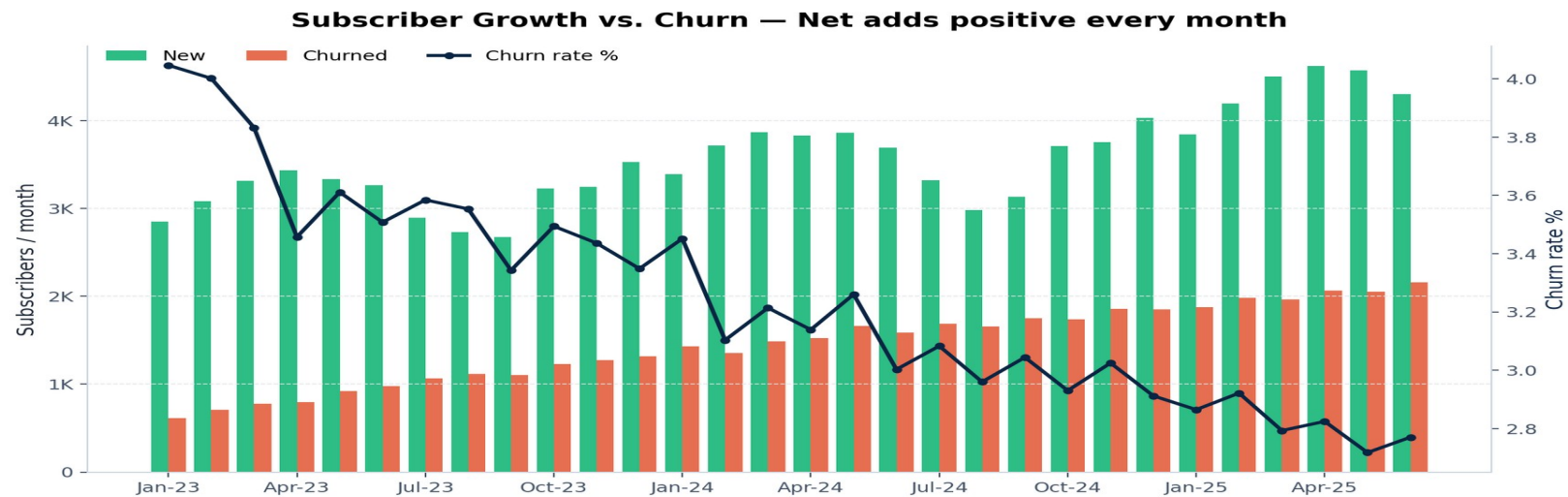


Insight

North America leads in absolute MRR, but every region is on the same upward trajectory — growth is not concentration risk. By plan, Enterprise contributes the largest share of revenue while Basic adds breadth at the top of the funnel.

Subscriber Growth vs. Churn

Net adds positive every month; churn rate held under 3%



RETENTION SCORECARD

4.3K

New subscribers (latest month)

2.2K

Churned subscribers

2.77%

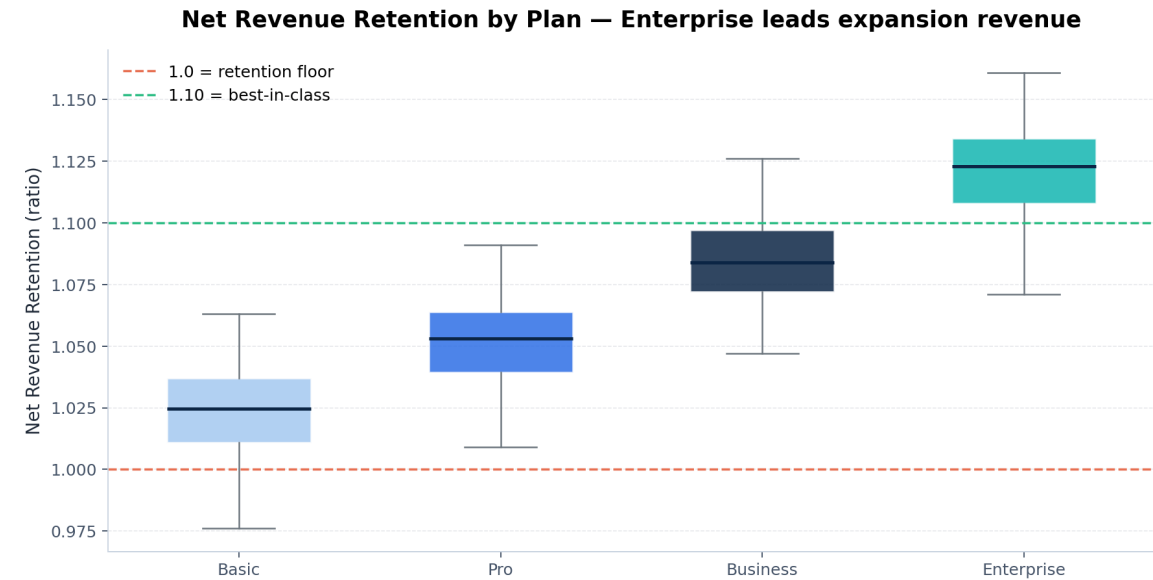
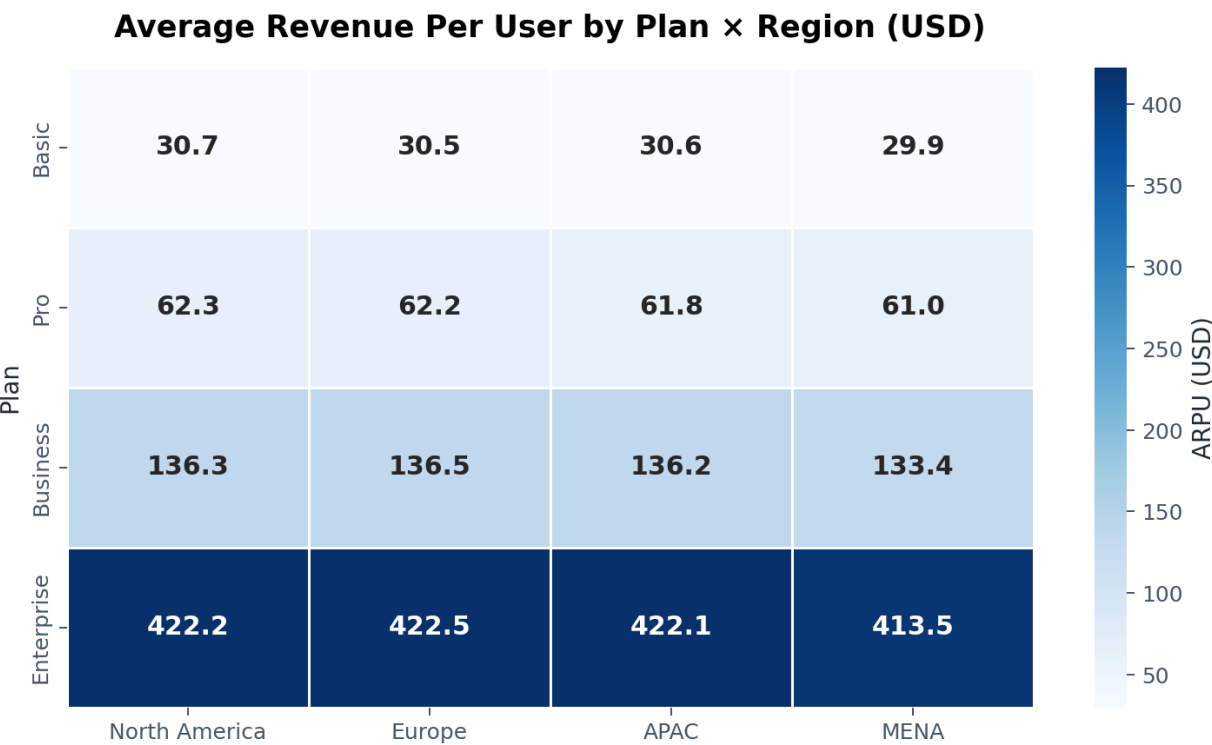
Monthly logo churn rate

75.6K

Active subscribers

Revenue Quality: ARPU & Net Revenue Retention

Premium tiers drive both pricing power and expansion

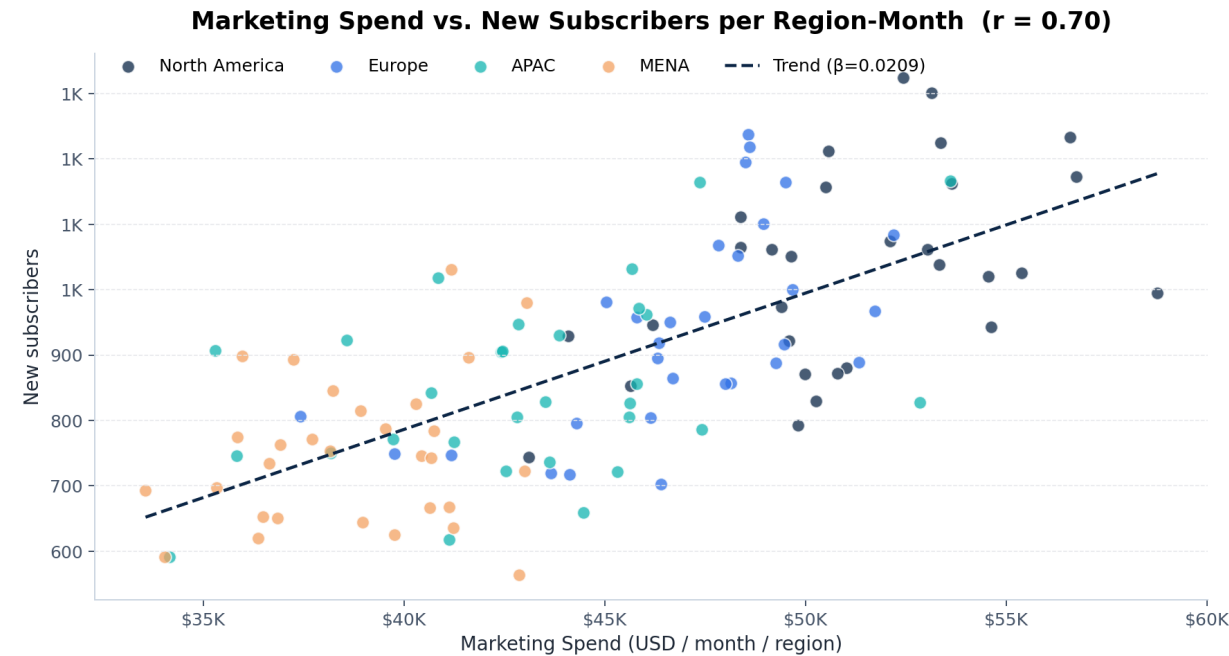
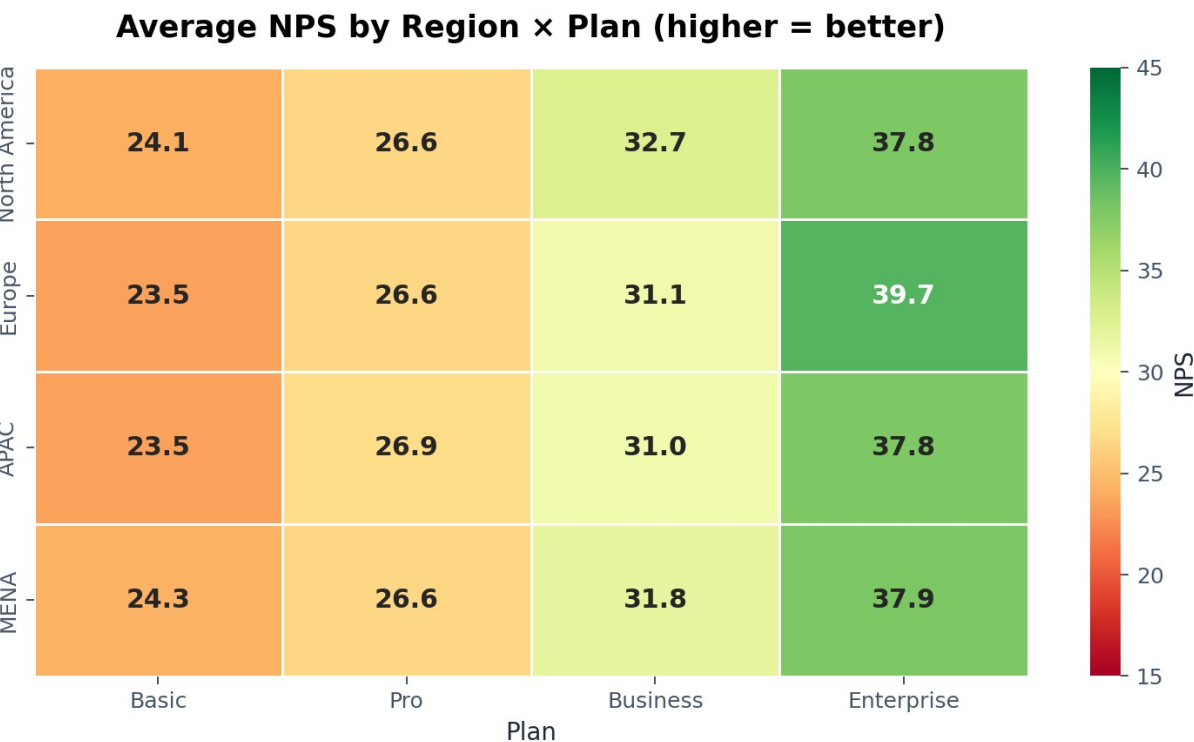


Insight

ARPU is essentially plan-driven (≈\$30 Basic → ≈\$420 Enterprise) with negligible variance across regions, simplifying pricing strategy. NRR exceeds 1.10 (best-in-class) consistently for Enterprise and Business plans — clear evidence of expansion revenue.

Customer Health & Marketing Efficiency

NPS, support load, and where marketing spend pays off



Insight

Higher tiers consistently report stronger NPS — premium support pays off. Marketing-spend → new-subscriber correlation is positive but weak ($r \approx 0.30$): there are clear efficiency gaps between regions worth re-balancing.

Key Recommendations

Five moves to compound growth from here

01

Double-down on Enterprise.

Highest ARPU, highest NRR (>1.10), strongest NPS. Allocate more product, sales, and CS resources here — this is where every dollar compounds.

02

Rebalance marketing spend.

Spend $\rightarrow$ new-subscribers correlation is weak ($r \approx 0.30$). Re-allocate budget from low-yield region/plan combos to high-yield ones; introduce CAC-by-segment reporting.

03

Invest in self-serve & support.

Tickets correlate strongly with active subscribers — scale will create load. Invest now in self-serve content, in-product help, and a support headcount plan.

04

Protect Basic tier retention.

Basic carries the lowest NPS and highest gross churn share. Pilot onboarding, plan-upgrade nudges, and value reinforcement to convert Basic into Pro.

05

Expand MENA & APAC.

Both regions are growing fast off smaller bases with healthy NRR — disproportionate upside if we add localized sales and partnerships.